

Healthy Skepticism · Session 3 take-home summary

Fault 3 is about the boxes your measurements get sorted into. Fault 4 is about the box a study result gets sorted into. Both take a continuous quantity, cut it in two, and then read the label as though it were the measurement.

Fault 3: arbitrary categories

Where does "high" begin?

Your systolic reads 120 this year. Last year it read 119. Nothing about you changed. The label did.

Every number a doctor measures is continuous. Every label is a box. Someone had to decide where the boxes begin, and the curve was not consulted.

One mmHg moved you across a line. It did not move your risk.

Systolic categories do not mean much

Plot ten-year cardiovascular risk against usual systolic pressure and it rises smoothly across every category boundary. Normal, Elevated, Stage 1 and Stage 2 are drawn on top of a curve that does nothing in particular at any of them.

There is a second lesson in the same chart. One cutoff, applied to a man and a woman, sits on top of very different absolute risks. The line is the same for both; what it means is not.

SCORE risk charts, low-risk regions: Conroy RM et al., Eur Heart J 2003;24:987-1003. Category boundaries: ACC/AHA 2017 hypertension guideline.

BMI categories are worse

The BMI case is stronger, because the boundaries fail in both directions at once.

- The 25 line, the start of "overweight", cuts straight through the flat basin of the curve. Crossing it takes the hazard ratio from about 1.02 to about 1.03.
- The 18.5 line already sits at 1.20, on a steeply rising limb. Underweight at BMI 16.7 carries a hazard ratio of 1.51, higher than anywhere in the entire "overweight" band. You do not meet that risk again until Obesity I, around BMI 33.
- Risk starts rising in the middle of "healthy weight", around BMI 21. The climb resumes in the middle of "overweight".

The curve is U-shaped. The risk transitions sit in the middle of the categories, and the boundaries sit where nothing happens.

Global BMI Mortality Collaboration, Lancet 2016;388:776-786. All-cause mortality hazard ratios relative to BMI 22.5-25, never-smokers, band-midpoint values with shape-preserving interpolation.

So why have categories at all?

Because a box is a decision rule. It tells a busy clinic who to look at twice, and it makes a chart readable in three seconds. That is a real use, and a good one.

The trouble starts when the box is read as a fact about your body. A box cannot say how much risk you carry, or how much of it you picked up by crossing the line.

Ask where you sit on the curve, not which side of the line you are on.

Fault 4: the Bernoulli fallacy

Findings that would not repeat

Only 6 of 53 landmark cancer findings were confirmed on rerun. Just 35 of 97 psychology findings stayed significant on rerun. This is the irreproducibility crisis, and the rest of this session is about where it comes from.

Begley CG and Ellis LM, Nature 2012;483:531-533. Open Science Collaboration, Science 2015;349:aac4716. Clayton A, Bernoulli's Fallacy, 2021.

An aside: conditional probability is the core element of inference

$P(A | B)$ is read as the probability of A if B is true, or more briefly, the probability of A given B. Depending on the circumstances, B may be observed, believed, or merely imagined for the sake of argument.

- **Courtrooms:** $P(\text{guilt} | \text{evidence})$
- **Gambling:** $P(\text{rolling 7 with two dice} | \text{the dice are fair})$
- **Science:** $P(\text{the tested hypothesis} | \text{the data})$
- **Causality:** $P(\text{drowning} | \text{warm weather})$

For this course you only need to be comfortable with the idea, not with any algebra.

Reversing a conditional probability

Day 2 did this twice. You can reverse $P(A | B)$ into $P(B | A)$ provided you also have $P(B)$, which is called the prevalence or the prior depending on the setting.

The mammogram is the worked case: $P(\text{called back} | \text{cancer})$ is 87%, the prevalence of cancer is about 6 in 1,000, and the two together gave $P(\text{cancer} | \text{called back}) = 4.4\%$.

Reversing a conditional probability using the prior is called Bayesian analysis.

Where the fallacy comes from

Researchers want to know whether a hypothesis is true. Rather than compute $P(\text{hypothesis} | \text{data})$, which is what they want, frequentist practice computes $p = P(\text{data} | \text{the hypothesis is false})$.

If p is small enough, the result is called statistically significant and submitted for publication. In medicine and the social sciences the criterion is p below 0.05: if the hypothesis were false, data like this would arise only 5% of the time. Some then treat the data as so surprising that the hypothesis can be taken as true.

That is Day 2's reversal again. And it carries an immediate consequence: on this criterion alone, about 1 in 20 published experiments would be expected not to reproduce.

The drawbacks

1. **The significance setting is arbitrary, and often far too high.** There is no formula or theorem that defines significance. 0.05 was offered as a convenience.
2. **Significance is not informative.** Knowing $P(\text{hypothesis} | \text{data})$ supports risk-based decisions, the way odds do in gambling. A p-value does not.
3. **It invites p-hacking.** Searching subsets of the data for one that meets the criterion by luck alone.

Together these feed the replication crisis.

Where the bar sits

Particle physics demands five sigma, a p-value below 0.000000287, one chance in 3.5 million. The Higgs waited for that bar. Medicine and the social sciences accept p below 0.05, one chance in twenty, which a test of nothing clears one time in twenty by luck.

Medicine cannot simply adopt five sigma: patients are scarce and trials are costly. But note what the 1 in 20 is. It is a rate per test of nothing. The share of published findings that are false is a different quantity, and getting to it is a reversal question: how many of the tested ideas were true to begin with? The prevalence, yet again.

Five sigma, one-sided, corresponds to $p = 0.000000287$; ATLAS and CMS required it for the 2012 Higgs discovery. Ioannidis JPA, PLoS Med 2005;2:e124.

So why adopt frequentism at all?

The honest case for it is that sometimes there is no prior to be had: industrial quality control, census survey sampling, particle physics. Where no prevalence is available, a method that does not require one is doing real work.

Its proponents also regard not using a prior as more objective and more honest. And there is a less flattering reason, which Clayton develops: a more rigorous standard would disrupt the academic publication world.

The excess is the evidence

Put the two numbers side by side. Honest testing at the 0.05 bar would produce a failure rate of about 1 in 20 on its own, with no torturing at all. What is observed is 47 of 53 landmark cancer findings failing to repeat, and about 2 in 3 in psychology.

The failure rate is far too high for luck alone. That gap is the evidence of p-hacking, also called data torturing.

Begley CG and Ellis LM, Nature 2012;483:531-533. Open Science Collaboration, Science 2015;349:aac4716.

Baby aspirin: the reversal patients lived through

In 1988 the Physicians' Health Study reported that aspirin cut first heart attacks by 44%, with p below 0.00001. It was stopped early for benefit, and decades of daily-aspirin advice followed.

In 2018 the retests arrived. ARRIVE reported $p = 0.60$. ASPREE found no net benefit and more major bleeding. In 2022 the USPSTF recommended against starting aspirin after age 60.

Trials stopped early tend to stop on a lucky high. On the retests, p never crossed the line again. The advice changed because the finding would not repeat, and the harm outweighed the benefit.

This concerns primary prevention. For people who have already had a heart attack or stroke the evidence is a different body of work and has not reversed. Do not stop a prescribed medication on the strength of a course handout.

Physicians' Health Study final report, N Engl J Med 1989;321:129-135. McNeil JJ et al. (ASPREE), N Engl J Med 2018;379:1509-1518. Gaziano JM et al. (ARRIVE), Lancet 2018;392:1036-1046. USPSTF, JAMA 2022;327:1577-1584.

From a p-value to policy

One 2010 paper: 31 autism cases, parent-reported rather than from medical records; roughly threefold odds, in boys only, not in the full sample; p below 0.05, so it was written up as a finding.

What followed is documented step by step.

4. One study: 31 parent-reported cases.

5. One number: p below 0.05.
6. One citation: the CDC's Autism and Vaccines page.
7. One committee: ACIP, twice, days before a vote.
8. One policy: universal newborn hepatitis B vaccination, ended.

The finding was never replicated, and in May 2026 the journal retracted the paper, months after the vote.

The retraction removed the paper from the literature. It did not restore the recommendation.

Gallagher CM and Goodman MS, J Toxicol Environ Health A 2010;73(24):1665-1677, since retracted. Large studies find no association: Price CS et al., Pediatrics 2010;126:656-664.

What to ask

- **Arbitrary categories.** What is my risk at my actual measurement, rather than which box it falls in? Where does the curve turn, and is it anywhere near the line?
- **The Bernoulli fallacy.** Have these studies been replicated? A single result clears the 0.05 bar by luck one time in twenty.
- How many analyses were run, and how many were reported?
- Was the trial stopped early, and on what?

The remaining fault is the causal leap: two things move together, and an arrow gets drawn between them without evidence that it belongs there.

Course materials and contact

Slides, videos and the whitepaper library are at healthy-skepticism.com. Decks live in the student area, behind the password given out in class.

Murray Cantor PhD, murray@murraycantor.com

Joel Keenan MD, drkeenan@mdvip.com